

Basic Machine Learning – I

Session Agenda

Duration: 3 Hours

Session Overview

This session builds the foundation of machine learning for beginners — starting with what ML is and how it differs from traditional programming, moving through data preprocessing (recap), train-test split, and model evaluation metrics, and then into hands-on regression and classification. Learners build and evaluate Linear Regression — including its core assumptions and hypothesis testing for coefficients — and Logistic Regression models, connecting each metric and algorithm to real predictive and classification use cases that underpin modern AI and Agentic AI systems.

Learning Outcomes

- Understand what Machine Learning is and its core types — supervised, unsupervised, and reinforcement learning
- Recall data preprocessing essentials (nulls, encoding, scaling) and perform a train-test split
- Interpret model evaluation metrics for classification (accuracy, precision, recall, F1, confusion matrix) and regression (RMSE, R^2)
- Build and interpret a Linear Regression model for continuous-value prediction
- Build and interpret a Logistic Regression model for binary classification
- Understand the key assumptions of Linear Regression (linearity, independence, homoscedasticity, normality of residuals, no multicollinearity) and interpret hypothesis testing for regression coefficients
- Apply evaluation metrics to compare model performance across algorithms

Detailed Session Schedule

Duration	Topic / Content
Part 1 — Foundations, Evaluation Metrics & Regression (90 min)	
10 min	Learning Outcomes: agenda walkthrough, and outcomes for the day.
15 min	Introduction to ML: What ML is, how it differs from traditional programming; types — supervised, unsupervised, reinforcement; where ML fits vs AI/DL/GenAI.
10 min	Data Preprocessing (Recap): Quick recap of handling nulls, categorical encoding, and feature scaling.
5 min	Train-Test Split: Why we split data into learning and checking parts, and the overfitting risk of skipping this step.
15 min	Model Evaluation Metrics: Accuracy, precision, recall, F1, and confusion matrix for classification; RMSE and R^2 for regression.
15 min	Practical 1: Preprocessing recap, train-test split, and computing a baseline evaluation metric in a beginner-friendly notebook.
20 min	Linear Regression: Cost function intuition, the best-fit line, and real-world use cases for predicting continuous values; core assumptions (linearity, independence, homoscedasticity, normality of residuals, no multicollinearity); and hypothesis testing for coefficients.

10 min	<i>Break</i>
Part 2 — Regression Practical & Logistic Regression (80 min)	
25 min	Practical 2: Linear Regression demo with scikit-learn — train a model, evaluate it using RMSE and R^2 , check assumptions via residual plots, and interpret hypothesis test output (p-values, confidence intervals) using statsmodels.
25 min	Logistic Regression: Sigmoid function, decision boundary, and why it is used for classification despite the name “regression.”
20 min	Practical 3: Logistic Regression demo — binary classification evaluated with accuracy, precision, recall, F1, and confusion matrix.
10 min	Recap + Q&A: Key takeaways from the session and open discussion.